

Website Traffic Analysis Report

1. Introduction

This project analyzes website traffic data to understand user behavior, top pages, referral sources, and user engagement. The analysis helps identify areas where Alfido Tech can improve website performance and increase conversions.

2. Data Cleaning & Preparation

The dataset was loaded into Google Colab using Pandas. The following preprocessing steps were performed:

- Removed missing/null values
- Checked duplicate records
- Converted date column into datetime format
- Standardized column names
- Prepared data for analysis and visualization

3. Key Metrics Computed

Total Events

Calculated total user interaction events from the dataset.

Total Users

Calculated unique users/links using available identifiers.

User Engagement

Analyzed user interactions based on event activity.

Monthly Traffic Activity

Measured traffic trends across different months.

4. Traffic Analysis

Top Countries

The majority of website traffic was generated from a few high-performing countries.

Top Cities

Certain cities showed significantly higher engagement and visits.

Artist-wise Traffic Analysis

Some artists received significantly more user interactions compared to others.

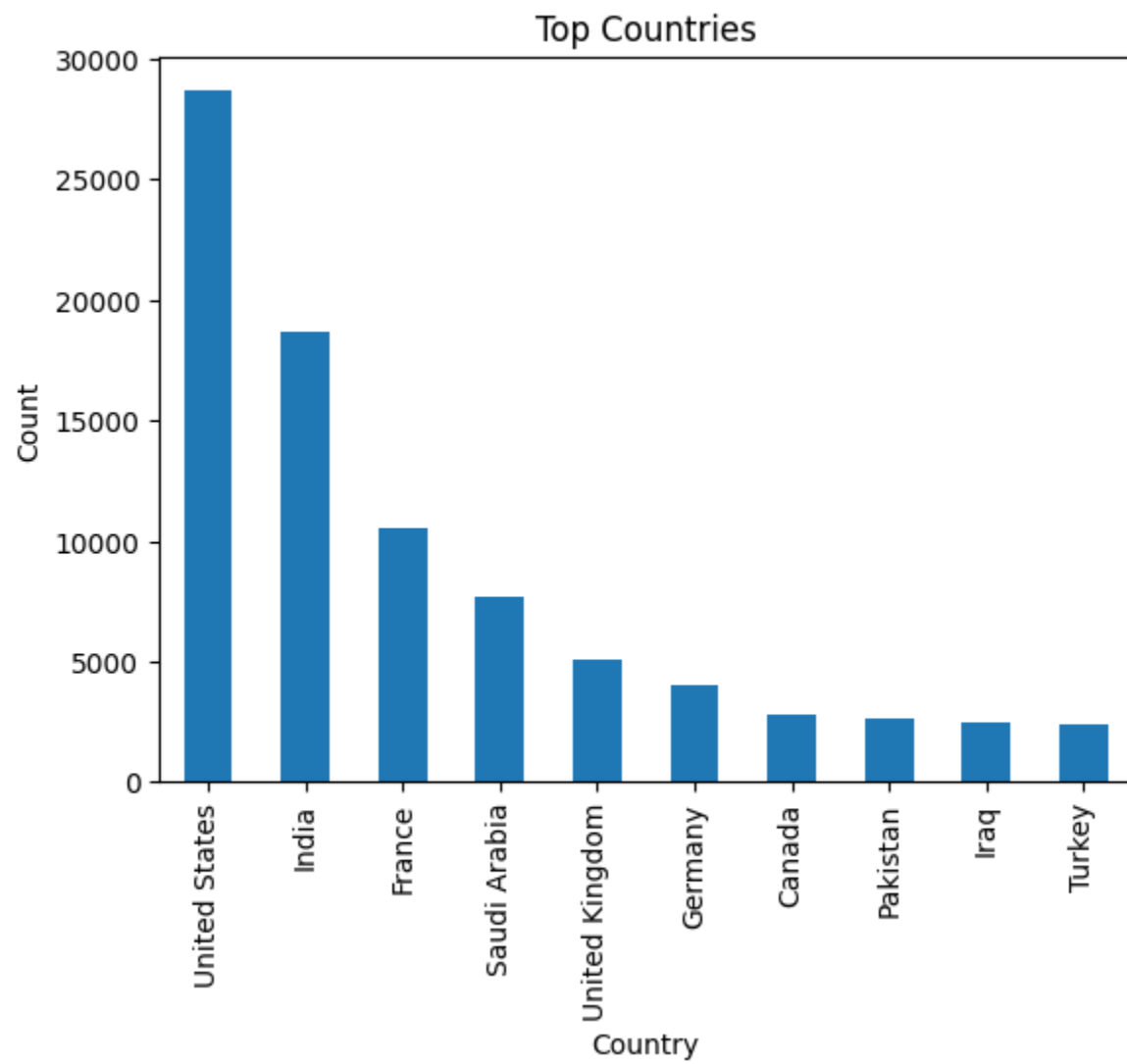
Album and Track Popularity

Certain albums and tracks generated higher user engagement and activity.

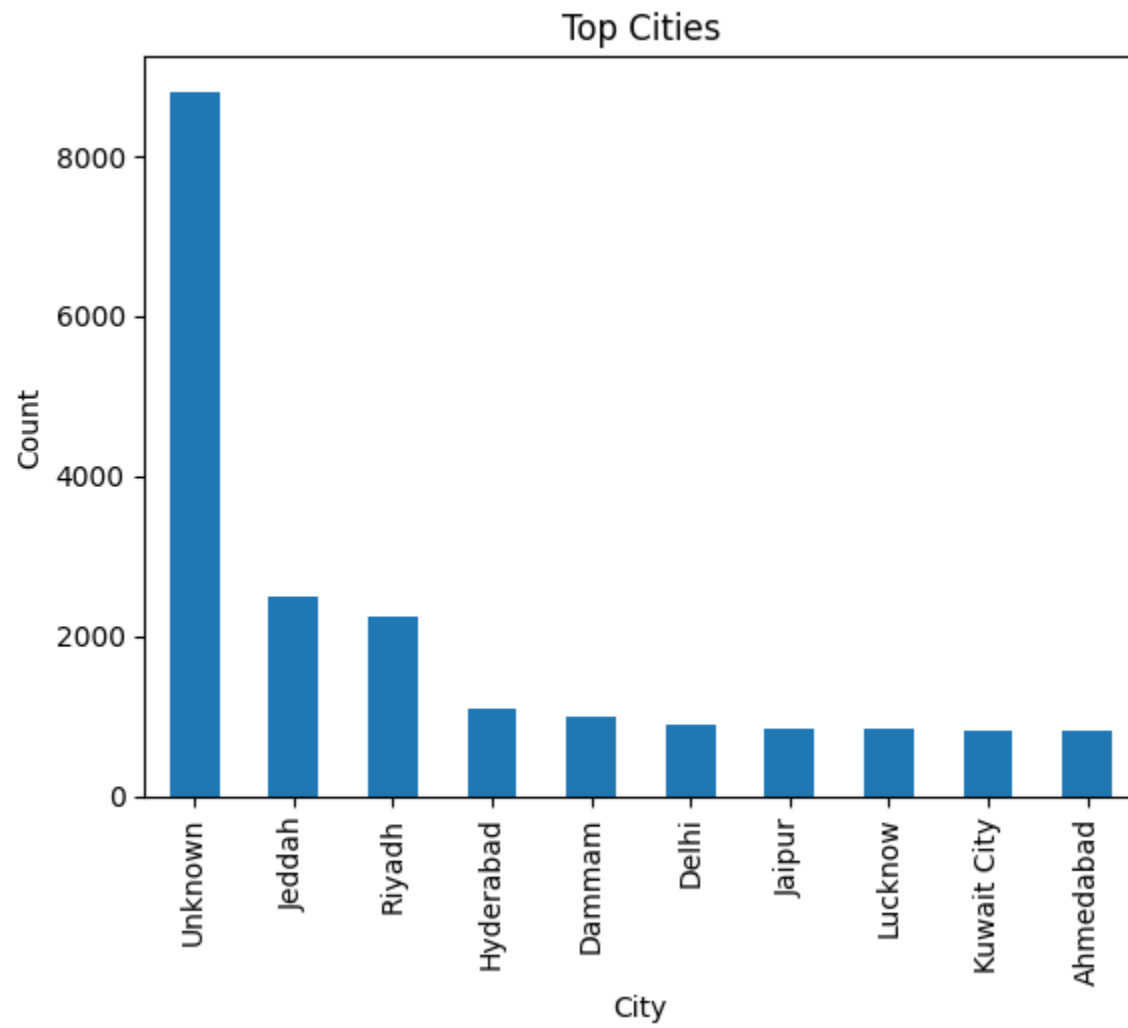
5. Visualizations Created

The following charts were generated:

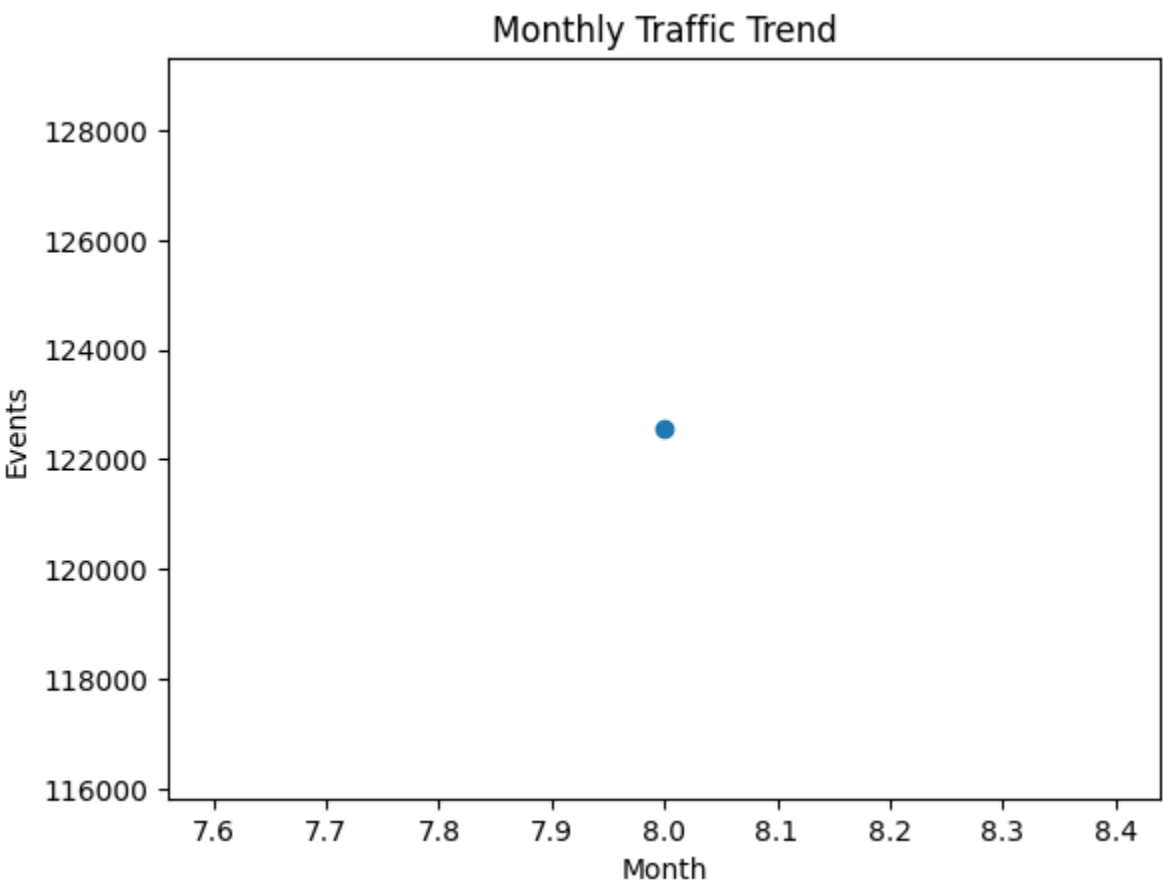
Top countries by traffic



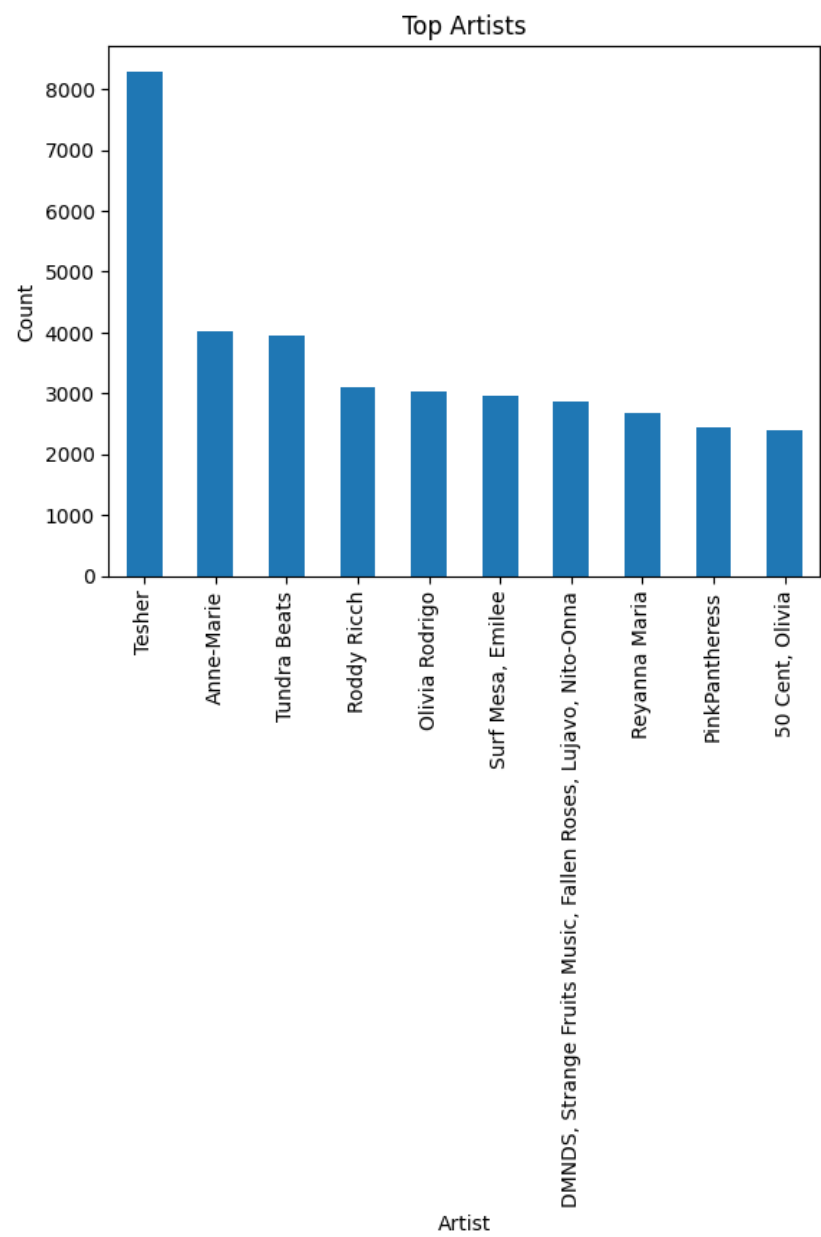
Top cities by traffic



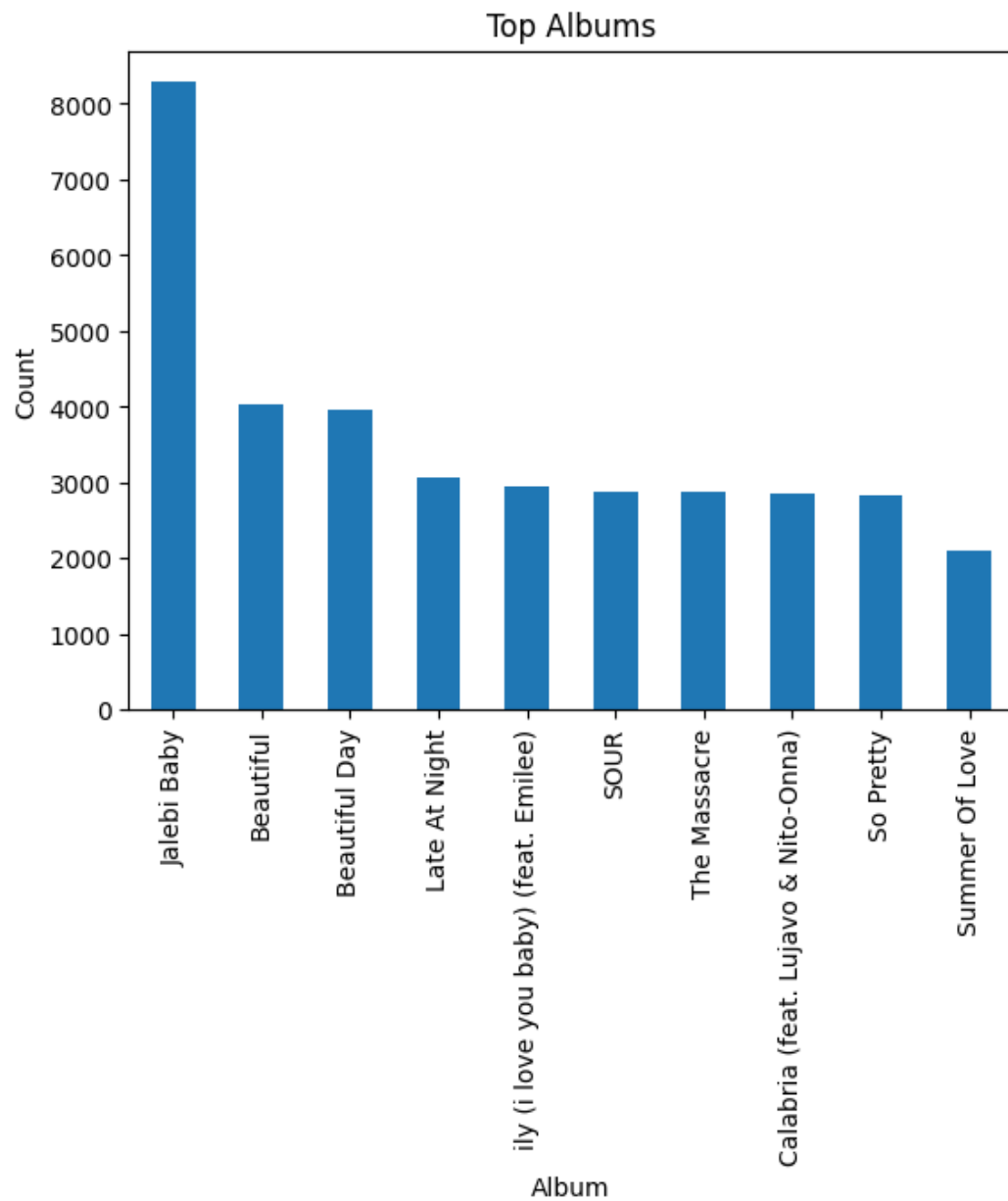
Monthly traffic trend



Artist-wise traffic analysis



Album/track popularity analysis



These visualizations helped identify user activity patterns and engagement trends.

6. Findings / Observations

- Website traffic was higher in certain countries and cities.
- Some artists and tracks received more interactions.
- Traffic trends changed across different months.
- User engagement varies across different content categories.
- Certain regions contributed more to overall website activity.

7. Recommendations

- Focus on promotions in high-traffic countries.
- Improve engagement strategies in low-performing cities.
- Promote trending artists and albums more effectively.
- Use seasonal marketing campaigns during high-traffic months.
- Improve user experience to increase user engagement.

8. Conclusion

The website traffic analysis provided valuable insights into user behavior, traffic distribution, and engagement patterns. The findings can help Alfido Tech improve website performance, increase user retention, and enhance conversion rates.